

```
170.1 171.7 172.6
>
```

```
> attributes(cpi)
$ts
[1] 1 12 1
```

```
$class
[1] "ts"
```

```
> typeof(cpi)
[1] "double"
```

The plot of the growing Consumer Price Index is shown in Figure 1.

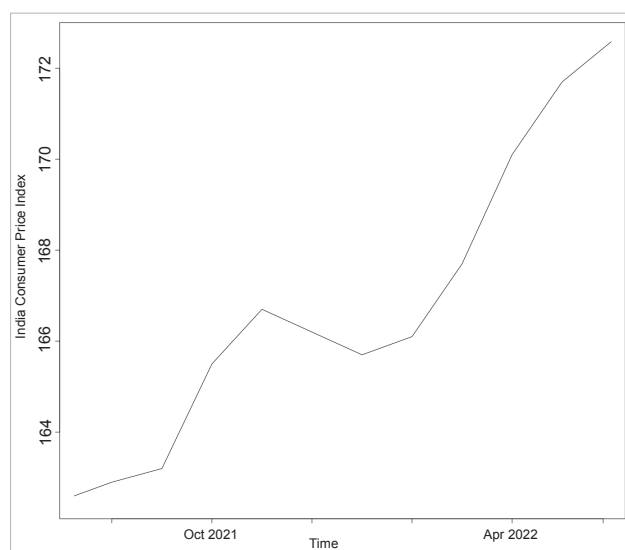


Figure 1: Consumer Price Index

The S3 classes do not have the structure of S3 objects and encapsulation is not enforced by the language. You can identify the hidden methods in a package by identifying an asterisk in their names, as shown below:

```
> methods(plot)
[1] plot,ANY-method      plot,color-method    plot.acf*
[4] plot.data.frame*     plot.decomposed.ts*  plot.default
[7] plot.dendrogram*     plot.density*        plot.ecdf
[10] plot.factor*         plot.formula*        plot.function
[13] plot.ggplot*         plot.gtable*         plot.hcl_
palettes*
[16] plot.hclust*         plot.histogram*      plot.
HoltWinters*
[19] plot.isoreg*         plot.lm*             plot.medpolish*
[22] plot.nlm*           plot.ppr*            plot.prcomp*
[25] plot.princomp*      plot.profile.nls*    plot.R6*
```

```
[28] plot.raster*      plot.shingle*      plot.spec*
[31] plot.stepfun      plot.stl*          plot.table*
[34] plot.trans*       plot.trellis*      plot.ts
[37] plot.tskernel*    plot.TukeyHSD*
```

You can also create an S4 class based on the S3 class using the `setOldClass` function. It accepts the following arguments:

Argument	Description
Classes	A character vector of old S3 classes
prototype	An object to use as a prototype
test	A logical value to test inheritance for the object
S4Class	A class name or definition for an S4 class

S4 class

The S4 class implementation is recommended for classes and methods for new software. The S4 classes support formal class definitions, multiple inheritance, encapsulation and parametric polymorphism. The `setClass()` function can be used to define a new class. We can define a class for *Car*, as shown below:

```
> setClass(
+   "Car",
+   representation(
+     mpg = "numeric",
+     cyl = "numeric",
+     disp = "numeric"
+   )
+ )
```

The following arguments are accepted by the `setClass()` function:

Argument	Description
Class	The name of the new class
representation	A list of slot and object names in the class
contains	A character vector of super classes
prototype	A default object for slots in the class
validity	A function to check validity of an object in the class
where	The environment to store the definition
sealed	A logical value that indicates if the class can be redefined
package	A package name for the class
S3 methods	A logical value on whether to have S3 methods for the class